

Quorum as a Where-Clause

Coalition Logic as a graded hypothesis language for populations of
computations,
with a swarm that decides and a pride that reads

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Abstract

A companion note read Gabbay’s Coalition Logic as a declarative specification language whose proofs, under a Curry–Howard term assignment, extract rho calculus implementations of consensus protocols. This note reads the same logic in the other direction, as the hypothesis language of a computation trying to predict a population of computations in its environment. The coalition modality $\exists\Box$ is placed in the *where*-clause of a rho for-comprehension and graded over a resolution semiring, joining the context-labelled behavioural modality $\langle K_j \rangle$ already used by the mortal scientist. The observational route needs only the model theory of Coalition Logic, which exists, and not its proof theory, which does not.

We work two examples. The first is a honeybee swarm choosing a nest site, where the observer is *inside* the population: a scout sensing a quorum at a cavity is evaluating the coalition modality about her own swarm, and her verdict is what triggers the swarm’s commitment. This case has four decades of measurement behind it, and the framework makes contact with it in three ways — it derives Seeley’s quorum-versus-consensus finding from an altitude argument rather than assuming it, it predicts a structural threshold at twice the quorum size across which the job of cross-inhibitory stop signalling changes from breaking deadlock to preventing splits, and it predicts that quorum-sensing errors are one-sided with the apparent threshold scaling in cavity geometry. The second example is a lion pride reading a stampeding herd, where the observer is *outside*; it is simulated rather than measured, and it isolates results the internal case cannot show: that the coalition modality buys irreversibility rather than accuracy, and that a population resists being read by denying its observer coverage rather than by inducing error — losing its own capacity to agree in the same move.

Across both: quorum intersection, the safety property of consensus, is what makes a population legible at all; and a mortal observer cannot distinguish a faulty participant from an observation it could not afford to finish.

Contents

1	Introduction	3
1.1	What is new here	4
1.2	Results	4
2	What is imported	5
Part I.	One modality, two directions	6

3	The reversal	6
3.1	Two readings of one table	6
3.2	What the observer does not need	7
4	Participants are process structure	8
5	The graded coalition modality	8
5.1	Grammar	8
5.2	The clause	9
6	Which algebra	9
7	Two bottoms	10
8	The semitopology is the hypothesis	11
8.1	Given versus guessed	11
8.2	The lattice of guesses	11
9	Legibility: what quorum intersection buys the observer	12
9.1	Measured	12
	Part II. The swarm: the observer inside	13
10	Nest-site selection as coalition detection	13
10.1	The position of this section	13
10.2	The mechanism, as measured	13
10.3	The mapping	14
11	Why quorum and not consensus	15
12	The $2k$ theorem and what stop signals are for	15
12.1	A structural bound	15
12.2	The regimes	16
12.3	Simulated	16
12.4	Disparity	16
13	One-sided errors and the encounter-rate prediction	17
14	What to measure	18
	Part III. The pride: the observer outside	19
15	The worked example: a pride and a herd	19
15.1	The population	19
15.2	The pride's ladder	19
15.3	The ladder measured	20
15.4	Lean versus commitment	21
15.5	Earliness	21
16	The counter-move	22
17	Compression: what a topen is worth	23

18 The margin is a clock	23
19 Altitude: where this sits in the choice hierarchy	24
19.1 Subject reduction, and the same suspect	25
20 What is shown and what is not	25
21 Open problems	26
22 Conclusion	27
A The pride and the herd in rholang	28
B The computations	30

1 Introduction

A swarm of honeybees hangs on a branch. Several hundred of its ten thousand workers are scouting the surrounding countryside for a cavity to live in, and over the next day or two the swarm will commit to one of them — irrevocably, and with no bee anywhere in possession of a global tally. What settles it, as Seeley and Visscher established, is not agreement among the dancers back at the cluster but a *quorum* sensed at a cavity: when enough scouts are present at one site, the scouts there return and pipe, the cluster warms up, and the swarm flies [SV04, See10].

A lion pride is stalking a herd. The herd startles, and in the next few seconds it will break — left or right, as one body or in fragments. The pride’s entire investment turns on committing to the correct flank before the break completes. It is not trying to *make* the herd break left; it is trying to *know* that the herd has already decided to.

These are the same question asked from two positions. In both, a computation wants to know whether a population of computations has reached a conclusion. In the swarm the asker is a member of the population and her verdict causes the conclusion to take effect; in the pride the asker is outside and can only watch. The first case has hard data behind it and is the centre of this note; the second is simulated, and earns its place by isolating things the first cannot show.

The companion note [Cons] observed that Gabbay’s Coalition Logic [GZ25, Gab25], interpreted over semitopologies [Gab23, Gab24], is a declarative language for distributed agreement, and proposed a Curry–Howard correspondence under which proofs of Coalition Logic formulae are decorated with rho calculus terms carrying join patterns, so that

$$\text{for}(y_1 \leftarrow x_{p_1} \ \& \ \cdots \ \& \ y_n \leftarrow x_{p_n}) P$$

witnesses the coalition modality $\exists\Box\varphi$. That is a synthesis programme: from specification to correct-by-construction implementation.

The present note runs the same correspondence backwards. Neither the scout nor the pride wants a term realising $\exists\Box\varphi$; each wants to evaluate whether $\exists\Box\varphi$ *holds* of a population it did not write. We are on the satisfaction side of the turnstile rather than the derivation side, and the computational vehicle is not a proof but a guard.

Three commitments organise the note.

The clause is the vehicle. Recent work [GW, ChF] developed the *graded where-clause*: the rho for-comprehension $\text{for}(y \leftarrow x \ \mathbf{where} \ \psi)$ whose guard is not a Boolean but a value in

a resolution semiring, keyed by spatial-behavioural formulae. A clause weighs the candidates in a fibre; it does not select from it. We add the coalition modality to the clause language. In the swarm the distinction between weighing and selecting is not a technicality but two different behaviours performed by two different sets of bees.

The semitopology is the hypothesis. In Gabbay’s setting a model *is* a semitopology (P, Open) : the collection of actionable coalitions is given with the model. Neither a scout nor a pride has such access. Their Open is a conjecture, and the relaxation lattice of [Sci] orders conjectures, so the machinery for revising a guessed semitopology is already built.

Everything is priced. The problem is not merely to be right but to be right *cheaply* and *early*. We therefore work in cost-accounted rho [CM], price each hypothesis, and report yields.

1.1 What is new here

Against [Cons]: the direction of the correspondence, the grading, the behavioural modality $\langle K_j \rangle$, the discovery of Open, and cost. Against [Sci, Play, Comp]: the object of study is a *population* rather than an individual, and the logic acquires a modality quantifying over coalitions rather than over redex positions. Against [GW, ChF]: a second modality in the same clause, and a use of the graded machinery which is observational rather than resolute — except in the swarm, where the two coincide, and which is therefore the first worked instance of the resolute reading those notes left open.

Against the honeybee literature the position is more delicate and is stated at the head of §10. There is a mature quantitative modelling tradition here — leaky competing accumulators, value-sensitive decision dynamics, best-of- n analyses [See12, Pai13, Rei17] — and nothing below out-predicts it on trajectories. What the framework contributes is a logic for *what* a scout is testing, a cost argument for why she tests that and not something else, and structural impossibility results that a statistical model is not in a position to state.

1.2 Results

- The satisfaction reading needs only the model theory of Coalition Logic. Problems P1–P5 of [Cons] — the missing sequent calculus, cut elimination, subject reduction for the term assignment, adequacy — are all on the synthesis side and are not obstacles here (§3).
- Section 7.1 of [Cons] contrasts decomposition over *participants* with decomposition over *process structure*. In a reflective calculus this contrast collapses: participants are processes, names are quotations of processes, and open-set quantification is separating conjunction at a different index (Proposition 6).
- Over $\mathbb{R}_{\geq 0}$ with multiplicative $\otimes$, larger coalitions score *weaker*, which is the wrong semantics for a quorum. The correct algebras are tropical and Viterbi, where the coalition value is the strength of the weakest member — and these are exactly the two algebras in which the attainment proposition of [ChF] holds on the nose, so the observer receives a *witness coalition* and not merely a number (§6).
- A mortal observer cannot distinguish Gabbay’s third truth value $\mathbf{b}$, a property of the participant, from its own exhausted budget, a property of the observation (Proposition 16).

- Quorum intersection — 2-twinedness, Gabbay’s *safety* condition — is what makes a population legible. Under it the population has at most one conclusion, so the observer’s hypothesis is a function (Proposition 23).

From the swarm (§10):

- Quorum-sensing at a site and consensus-sensing at the cluster are different formulae at different altitudes: the first is a local $\oplus \otimes$ over one open, the second a global universal. No bee can evaluate a global universal within its access grades, so the framework *derives* the finding of [SV04] (Proposition 33).
- **The $2k$ theorem.** Two simultaneous quorums require $2k$ committed scouts, so a swarm whose active scout pool falls short of twice the quorum threshold cannot split, whatever the site qualities and whatever the stop signalling (Theorem 36). Measured: splits are identically 0.0000 at every $S < 30$ and first appear at $S = 30$ with $k = 15$.
- **Cross-inhibition changes jobs at $S = 2k$.** Below the threshold the failure mode is deadlock and stop signalling removes it entirely ($0.5120 \rightarrow 0.0000$ at $S = 16$); above it the failure mode is splitting and stop signalling only mitigates it ($0.3860 \rightarrow 0.1170$ at $S = 40$). This is a discriminating prediction: the two accounts of what stop signals are for should be separable by manipulating the scout pool (§12).
- **Quorum errors are one-sided, and the threshold is not a number.** If a scout senses quorum by encounter rate, false positives are structurally impossible and the *apparent* threshold in bees scales linearly with cavity geometry: measured 16, 21, 25, 33 true bees for effective areas 1, 1.25, 1.5, 2 against a nominal $k = 15$ (§13).

From the pride (§15–§17):

- The coalition modality does not buy accuracy; it buys *irreversibility*. Over 6839 coalition verdicts the precision was 1.0000 at every observation time, while a five-animal sample ranged from 0.9160 to 0.9873. The cost of the coalition rung is $19\times$ the sample’s and its folded accuracy is *lower* (§15.3).
- A population defends itself against being read not by deceiving the observer but by denying it coverage. As the herd’s internal seams grow from 0 to 6, the pride’s coverage falls from 0.7520 to 0.0115 while its precision stays at 1.0000 — safety preserved, liveness lost. The herd pays in its own consensus, which falls from 0.7257 to 0.2331 (§16).
- Grading at the coalition size is crisp in disguise: min over thirteen animals is zeroed by one dissenter, and the crisp and graded rungs return identical verdicts. The grading pays only when the coalition size is *relaxed* at the same time — at which point the margin becomes a clock. Both examples show this, with waits falling monotonically across margin quartiles: 42.4, 29.6, 28.9, 18.0 ticks for the herd and 59.7, 35.1, 18.4, 8.4 for the swarm (§18).

Every number is computed by `swarm.py` and `herd.py`, described in Appendix B.

2 What is imported

We assume, and do not restate, the following.

From [Cons]. Semitopologies (P, Open) , where Open contains $\emptyset$ and P and is closed under arbitrary unions but *not* under finite intersection; points are participants and opens are actionable coalitions. Semiframes and the compatibility relation $*$. Coalition Logic: a three-valued modal fixed-point logic over $\mathbf{3} = \{\mathbf{f} < \mathbf{b} < \mathbf{t}\}$, whose central modality is

$$\llbracket \exists \square \varphi \rrbracket(p) = \mathbf{t} \quad \text{iff} \quad \exists O \in \text{Open}. p \in O \wedge \forall q \in O. \llbracket \varphi \rrbracket(q) = \mathbf{t}.$$

n -twinedness (every n nonempty opens meet), topens (transitive opens, whose members are guaranteed to decide alike), and the classification of points.

From [Sci, Play]. The mortal scientist: a cost-accounted rho computation whose environment is other computations, separated from it by a cut across which there are three grades of access — perception, interaction, reflection. The namespace logic: one structural connective per term former, including separating conjunction $\varphi \mid \psi$ and the name predicate $@\varphi$; one primitive behavioural modality $\langle K_j \rangle$ per rewrite rule *and* choice of redex position, with $\exists j. \langle K_j \rangle \varphi$ and $\forall j. [K_j] \varphi$ quantifying over redex positions; the greatest fixed point $\nu X. \varphi$; graded separation $\varphi \mid_k \psi$, regaded over a described namespace as $\varphi \mid_k^N \psi$ once the hidden-name quantifier is retired by freshness-by-quotation. Depth grading $d(\cdot)$ and the pricing schedule $\kappa(d) = 2^d$ per candidate examined. The assay trichotomy: confirm, refute, or $\perp$, where $\perp$ is budget-relative. The read-versus-drive thesis: a predicate may be *read* from structure the environment has already computed into legible form, priced by formula depth, or *driven* by interaction, priced by rendezvous.

From [Comp]. A learner is a distribution over a population of computations, and that population constitutes a *namespace*. Learners compose by parallel composition; overlap of namespaces is the dial between non-interaction, competition and cooperation.

From [GW, ChF]. The graded where-clause: the guard of a for-comprehension evaluates into a commutative resolution semiring $(V, \oplus, \otimes, 0, 1)$ rather than into Booleans, with the graded formula sort

$$\psi ::= [\beta] \mid v \mid \psi \otimes \psi \mid \psi \oplus \psi \mid \langle K \rangle \psi \mid X \mid \mu X. \psi \mid \nu X. \psi$$

($[\beta]$ the embedding of a crisp formula, $v \in V$ a constant), the algebra table — $\mathbb{B}$, $\mathbb{R}_{\geq 0}$, $\mathbb{C}$, Viterbi $(\max, \times)$, tropical $(\min, +)$ — and the attainment proposition: with idempotent $\oplus$ and crisp ψ a definable section exists; with idempotent $\oplus$ and graded ψ the value is attained exactly when the winning candidate carries the multiplicative unit; with non-idempotent $\oplus$ the clause is an expectation and is never attained into the fibre. Tropical and Viterbi are the two rows where selection is definable outright. Also: the division of labour between clause, choice principle and scheduler — the clause is a *weighting* of the fibre, a quantifier in the Escardó–Oliva K -monad; a scheduler is a *section*, a selection function in the J -monad.

Part I. One modality, two directions

3 The reversal

3.1 Two readings of one table

The correspondence table of [Cons, §5.2] reads, in part,

Coalition Logic	Rho with joins
participant p	channel name x_p
open $O = \{p_1, \dots, p_n\}$	channel set $\{x_{p_1}, \dots, x_{p_n}\}$
$\exists \Box \varphi$ at p	$\text{for}(y_1 \leftarrow x_{p_1} \ \& \dots \ \& y_n \leftarrow x_{p_n}) P_\varphi$
t	running term
b	blocked or stuck term (faulty participant)
f	dead term

and its rules are decorations: $\exists \Box R$ introduces the join, $\exists \Box L$ introduces a send by which a participant contributes local evidence, cut elimination is the message passing. This is a term assignment, $\vdash \varphi \triangleright P$: a formula's realiser.

The pride needs the other judgement.

Definition 1 (The two readings). Let φ be a formula of Coalition Logic and P a rho process.

- The *synthesis* reading is the term assignment $\vdash \varphi \triangleright P$: from a proof of φ , extract P .
- The *observation* reading is the satisfaction relation $P \models \varphi$: given P , decide φ .

Proposition 2 (The clause is the observation side). A where-clause is an occurrence of the observation reading. Its guard is evaluated against a candidate drawn from the store; the guard's verdict admits or refuses the candidate; the guard constructs nothing.

Proof. Immediate from the operational semantics of the clause [GW]: for a comprehension $\text{for}(y \leftarrow x \text{ where } \psi) P$ and a candidate datum d on x , reduction is licensed by $\llbracket \psi \rrbracket(d) \neq 0$ and the reduct is $P[d/y]$; ψ does not appear in the reduct. $\square$

Proposition 2 is small but it settles the shape of everything that follows. It also explains why the resolute reading of consensus — using agreement among a population to settle a race in the observer — is the strained direction, even though it is the one that first suggests itself.

Remark 3 (Weighing is not selecting). In the vocabulary of [ChF], a graded clause is a map $\text{Cand}(P, r) \rightarrow V$: a weighting of the fibre of candidates, hence a quantifier in the K -monad $(X \rightarrow R) \rightarrow R$. A resolver is a section, an element of the J -monad $(X \rightarrow R) \rightarrow X$. The pride is a K -monad user. It evaluates how firmly a coalition has converged without selecting a member of it, and would not be helped by a selection: it does not get to pick which zebra runs where. This is a type-level statement of the ecological fact.

3.2 What the observer does not need

Proposition 4 (The observation reading is proof-theory-free). The observation reading of Coalition Logic requires only the model theory of [GZ25]. In particular it does not require: a sequent calculus for Coalition Logic (P1 of [Cons]); a geometric characterisation of the well-behavedness conditions in Negri's sense (P2); a coalgebraic account of the modality supporting generic cut elimination (P3); a term assignment satisfying subject reduction and progress (P4); or an adequacy theorem for extracted terms (P5).

Proof. The observation reading evaluates $\llbracket \varphi \rrbracket(p)$ directly from the semantic clauses. Each of P1–P5 constrains the passage from proofs to terms; no such passage is taken. $\square$

This is the strategic content of the note. The companion programme is blocked on a piece of proof theory that nobody has written. The observer programme is not blocked on anything: Gabbay's semantics is complete as it stands, and what remains is to price it and to grade it.

Remark 5 (What is lost). Correspondingly, the observer gets no correctness-by-construction. Its verdicts are as good as its guessed Open, which is the whole subject of §8. The two programmes are not competitors; they occupy opposite sides of a turnstile whose two halves are, as always, of very different difficulty.

4 Participants are process structure

Section 7.1 of [Cons] notes the structural resemblance between the coalition modality and the spatial modalities of Caires and Cardelli [CC01, CC04], and records a difference: Coalition Logic’s open-set quantification decomposes over *participants* where spatial logic’s $\varphi \mid \psi$ decomposes over *process structure*. A systematic comparison, the note says, may yield insights in both directions. In a reflective calculus the difference is not there to be compared.

Proposition 6 (Collapse). *Let P be a rho process presented as a parallel composition $P \equiv \prod_{i \in I} P_i$ of participant processes, with participant i addressed at the manufactured name $x_i = @C_i[P_i]$ for a context C_i properly containing P_i . Then quantification over subsets $O \subseteq I$ of participants and decomposition of P by separating conjunction are the same operation, indexed differently: for any predicate φ ,*

$$\exists O \subseteq I. (\forall q \in O. P_q \models \varphi) \quad \text{iff} \quad P \models \left(\left(\bigvee_{q \in O} \varphi \right) \mid \top \right) \text{ for some } O,$$

and the namespace $\{x_q : q \in O\}$ is definable by a name predicate $@\varphi_O$.

Proof. Participants are processes, so the collection $\{P_q\}_{q \in O}$ is a sub-multiset of the parallel decomposition of P ; separating conjunction asserts exactly the existence of such a decomposition. Conversely, names are quotations of processes, so a set of participant addresses is a set of names, and by freshness-by-quotation [Sci, §3] the address of q is a term-level function of P_q ; a predicate cutting out O at the process level therefore cuts out $\{x_q\}_{q \in O}$ at the name level via $@$. The two quantifications range over images of one another under quotation. $\square$

Corollary 7 (Opens are described namespaces). *A guessed semitopology on a population P is presented by a family of name predicates: $O \in \text{Open}$ is given as $@\varphi_O$, and the closure of Open under arbitrary unions is the closure of that family under disjunction of name predicates.*

Corollary 7 is what makes the whole construction expressible in the logic already available to the mortal scientist. It also imports the right grading apparatus for free: graded separation over a described namespace, $\varphi \mid_k^{\mathcal{N}} \psi$, is a statement about how far apart two sub-populations are within a named coalition structure.

Remark 8 (What does *not* collapse). The absence of a meet is untouched by Proposition 6. Spatial logic’s $\mid$ is an operation on decompositions; Open is not closed under intersection, and no amount of reflection makes it so. The collapse is of the *index*, not of the algebra: participants and process structure are one domain, but the collection of *actionable* sub-populations remains a join-semilattice with a compatibility relation, not a lattice.

5 The graded coalition modality

5.1 Grammar

Fix a commutative resolution semiring $(V, \oplus, \otimes, 0, 1)$ and a population namespace $\mathcal{N}$ carrying a guessed Open presented as in Corollary 7. Extend the graded sort of [GW, ChF] with one clause.

Definition 9 (Graded coalition modality).

$$\psi ::= [\beta] \mid v \mid \psi \otimes \psi \mid \psi \oplus \psi \mid \langle K \rangle \psi \mid \langle E \rangle_V^N \psi \mid X \mid \mu X. \psi \mid \nu X. \psi$$

with

$$\llbracket \langle E \rangle_V^N \psi \rrbracket(p) = \bigoplus_{\substack{O \in \text{Open} \\ p \in O}} \bigotimes_{q \in O} \llbracket \psi \rrbracket(q).$$

Taking $V = \mathbb{B}$ with $\oplus = \vee$, $\otimes = \wedge$ recovers Gabbay's clause exactly: the $\bigoplus$ is the $\exists O$, the $\bigotimes$ is the $\forall q \in O$. So Definition 9 is a grading of $\exists \Box$ and not a new modality.

Remark 10 (Two modalities, one clause). $\langle K_j \rangle$ and $\langle E \rangle_V^N$ index different things: $\langle K_j \rangle$ ranges over redex positions — one step of the population's own computation — and $\langle E \rangle_V^N$ ranges over actionable sub-populations. They live in the same clause and nest freely. The formula that says the interesting thing is their interleaving under a greatest fixed point:

$$\Phi_{\text{coh}} := \nu X. \langle E \rangle_V^N ([\beta] \otimes \langle K \rangle X)$$

— *some coalition agrees on β , and continues to agree as the population steps*. This is herd coherence, and it is the formula a pride would most like to be able to afford.

Remark 11 (Why the fixed point must be graded). Φ_{coh} is not expressible in the graded sort as originally presented in [GW], which deliberately excluded fixed points. It became available with the graded μ and ν of [ChF], whose existence conditions are algebra-dependent: Knaster–Tarski for a quantale, convergence for $\mathbb{R}_{\geq 0}$, a spectral condition for $\mathbb{C}$. Tropical and Viterbi, which §6 selects, are idempotent and bounded, so ν exists by Knaster–Tarski on the value order.

5.2 The clause

The observer's hypothesis is carried in a guard:

```
for ( @sighting <- senses where <E>_N ( heading == d ) ) { commit!(d) }
```

The pride's commitment fires when, and only when, the graded coalition value clears zero. The verdict is three-valued in the sense that matters operationally: it fires for d , it fires for the opposite of d , or it does not fire.

6 Which algebra

The grammar of Definition 9 is parametric in V . The choice is not free, and getting it wrong produces a modality that says the opposite of what a quorum means.

Proposition 12 (Large-coalition decay). *Let $V = \mathbb{R}_{\geq 0}$ with $\otimes = \times$, and suppose $\llbracket \psi \rrbracket(q) \leq c < 1$ for all q . Then for opens $O \subsetneq O'$ with $|O'| = |O| + m$,*

$$\bigotimes_{q \in O'} \llbracket \psi \rrbracket(q) \leq c^m \bigotimes_{q \in O} \llbracket \psi \rrbracket(q),$$

so the value of a coalition decays geometrically in its size, and $\langle E \rangle_{\mathbb{R}_{\geq 0}}^N$ is maximised by the smallest admissible coalition.

Proof. Immediate. □

This inverts the intended reading. A quorum is more compelling when larger, not less; a modality that prefers the minimal open reports the opposite of agreement.

Definition 13 (The quorum algebras). Take V idempotent with $\otimes = \min$: either Viterbi ($\max, \min$) on $[0, 1]$, or tropical ($\min, +$) read as cost with the order reversed. Then

$$\llbracket \langle E \rangle_V^N \psi \rrbracket(p) = \max_{O \ni p} \min_{q \in O} \llbracket \psi \rrbracket(q) :$$

a coalition is as strong as its weakest member, and among coalitions the observer reports the strongest.

Proposition 14 (The observer gets a witness). *Under Definition 13, $\langle E \rangle_V^N$ is attained: there is a definable function returning an $O^* \in \text{Open}$ realising the value.*

Proof. $\oplus = \max$ is idempotent and the index set of opens is presented, so the first two clauses of the attainment proposition of [ChF] apply; tropical and Viterbi are exactly the rows in which the selection is definable outright rather than merely existing. $\square$

Proposition 14 is what makes the machinery ecologically usable. A number is not actionable; a pride cannot charge at a scalar. The witness O^* is a *located* sub-population — these zebra, over there — and the flank is chosen by its position. It is worth recording that the ecological reading lands, without being steered, in exactly the fragment of the choice hierarchy where selection is constructively available.

Remark 15 (The cost of min: grading at the coalition size is crisp in disguise). $\min$ is brittle. One member with $\llbracket \psi \rrbracket(q) = 0$ zeroes the whole coalition, so when the coalition is large the graded modality reproduces the crisp one. This is not a conjecture: in the worked example at $k = 13$ the crisp and graded rungs return identical verdicts on every trial and their measured coverage and precision agree to four places (Table 4). The grading pays only when the *coalition size is relaxed at the same time*, which is to say when the observer descends the relaxation lattice as well as changing the algebra. Two dials, and they must be turned together.

7 Two bottoms

Gabbay's third truth value **b** is a property of a participant: faulty, Byzantine, stuck. The mortal scientist's $\perp$ is a property of an observation: the assay was not completed within budget [Sci, §4]. These are different things, and the note's least comfortable result is that the observer cannot tell them apart.

Proposition 16 (Fault and poverty are one value). *Let S be a mortal observer with purse σ evaluating $\langle E \rangle_V^N \psi$ at p over a population E , and let the evaluation be priced per participant examined. If the full evaluation costs more than σ , then S 's verdict is **b**. This verdict is indistinguishable, from within S , from the verdict **b** returned when the evaluation completes and some $q \in O$ is itself **b** for every $O \ni p$.*

Proof. The two verdicts are the same element of **3**, and distinguishing them would require either completing the evaluation — which is what σ forbids — or an oracle across the cut reporting the reason for non-completion, which the three grades of access do not provide [Sci, §2]. $\square$

Corollary 17 (Poverty manufactures faults). *An impoverished observer perceives a population as more faulty than it is, with the error one-sided: it never perceives a fault where there is none in the observer's own dynamics, but it systematically over-reports.*

purse	$\Pr(\mathbf{b})$	of which decided if rich
60	0.6104	0.5668
120	0.3104	0.1482
200	0.2644	0.0000
320	0.2644	0.0000
480	0.2644	0.0000

Table 1: Two bottoms. At a purse of 60 the observer returns $\mathbf{b}$ on 61.0% of trials and 56.7% of those are its own poverty rather than the herd’s indeterminacy. Above a purse of 200 the confusion vanishes and the residual 26.4% is the herd genuinely not having decided. 2500 trials.

This is the over-time clause of the modal asymmetry of [Sci, Prop. 6] landing in a new place. There it said that a step not yet enabled is indistinguishable from a step never enabled. Here it says that a participant not yet examined is indistinguishable from a participant that is broken.

The effect is measurable. Table 1 gives, for the worked example at $k = 13$ and observation time $t = 90$, the probability that the observer’s verdict is $\mathbf{b}$ as a function of its purse, and the fraction of those $\mathbf{b}$ verdicts that a purse-unconstrained observer would have decided.

Remark 18 (A design consequence). Since the two bottoms are one value, an observer that wishes to distinguish them must *instrument itself*: it must record, alongside the verdict, the budget consumed and whether the scan terminated. That record is not part of the logic and cannot be, because it is a fact about the observer rather than about the population. This is a small concrete instance of the cut doing work: the logic describes the far side, the accounting describes the near side, and neither can be reduced to the other.

8 The semitopology is the hypothesis

8.1 Given versus guessed

In [GZ25, Cons] a model is a semitopology: Open arrives with the model, and correctness arguments quantify over the opens as given data. This is appropriate for a protocol designer, who chooses the quorum system. It is not available to an observer.

Principle 19 (The observer’s inversion). For a mortal scientist reading a population, Open is not part of the model; it is the content of the hypothesis. A theory of a population is a guess at which sub-populations can act as one.

Principle 19 is what turns Coalition Logic from a specification language into a hypothesis language, and it is the reason this note exists as something other than a restatement of [Cons].

8.2 The lattice of guesses

The apparatus for ordering guesses is already built. In [Sci, §5–6] hypotheses live in a relaxation lattice ordered by logical strength, and the ν -stratified ultrametric puts a tree geometry on them: balls are nested or disjoint, every point is a centre, there is no betweenness and hence no gradient. Guessed semitopologies inherit this.

Definition 20 (Refinement of guesses). Guesses at Open are presented by families of name predicates (Corollary 7). $\text{Open}_1 \preceq \text{Open}_2$ when every open of Open_1 is an open of Open_2 .

Descending the order — *coarsening*, requiring larger coalitions — strengthens the hypothesis and raises the cost of confirming it; ascending — *relaxing* — weakens it and lowers the cost.

Proposition 21 (Coarsening costs, twice). *Let the basis of Open be the sub-populations of size at least k . Then increasing k both raises the cost of evaluating $\langle E \rangle_V^N$, since each open tested examines at least k participants, and lowers its coverage, since fewer states admit a uniform open of size k .*

Both effects are measured in §9 (Table 2): as k rises from 4 to 24 the probability that neither direction is decided rises from 0.0000 to 0.9252.

Remark 22 (No gradient here either). Because the geometry is ultrametric, an observer that only ever relaxes or coarsens by one step cannot escape the ball it started in [Sci, Prop. Confinement]. A guess at Open that is wrong in its *shape* — the herd is two clusters and the observer believes it is a ring — is not reachable from a guess that is wrong only in its size, at any budget. Recombination, not refinement, is what crosses that distance.

9 Legibility: what quorum intersection buys the observer

Gabbay’s n -twinedness generalises quorum intersection: every n nonempty opens meet. In consensus theory this is a *safety* condition — it is what forbids two quorums deciding differently. Read from outside, it does something else.

Proposition 23 (Twinedness makes the hypothesis functional). *Let (P, Open) be 2-twined and let $\beta_1, \dots, \beta_m$ be pairwise exclusive crisp predicates on participants. Then at most one i has $\llbracket \exists \square \beta_i \rrbracket(p) = \mathbf{t}$.*

Proof. Suppose $\llbracket \exists \square \beta_1 \rrbracket(p) = \llbracket \exists \square \beta_2 \rrbracket(p) = \mathbf{t}$ with witnessing opens O_1, O_2 . By 2-twinedness $O_1 \cap O_2 \neq \emptyset$; take q in the intersection. Then $\llbracket \beta_1 \rrbracket(q) = \llbracket \beta_2 \rrbracket(q) = \mathbf{t}$, contradicting exclusivity. $\square$

Corollary 24 (Legibility). *Under 2-twinedness the map “which conclusion has this population reached?” is a partial function. Without it, it is a relation, and an observer that must commit to one answer has no well-posed question to ask.*

Principle 25 (Safety is legibility). The property that makes a distributed system safe from the inside is the property that makes it readable from the outside. They are the same proposition, evaluated by different parties.

This is the note’s first substantial payoff from the change of direction, and it is not visible from the synthesis side, where quorum intersection appears only as a hypothesis of a correctness proof.

9.1 Measured

In the worked example (§15) the population is a ring of $N = 24$ and the basis of Open is the set of contiguous arcs of length at least k . Two arcs of length k overlapping in fewer than k points meet in a non-open set, so this is a semitopology and not a topology. On a ring, 2-twinedness holds exactly when $k > N/2$.

The distinction between $k = 12$ and $k = 13$ in Table 2 is worth dwelling on, because both rows read 0.0000 and they mean different things.

Observation 26 (Statistical zero and structural zero). At $k = 12$ two disjoint decided coalitions are possible: the state consisting of twelve contiguous participants heading left and twelve heading right has $\llbracket \exists \square \beta_L \rrbracket = \llbracket \exists \square \beta_R \rrbracket = \mathbf{t}$. Verified by construction. At $k = 13$ no pair

k	both decided	one decided	neither
4	0.0944	0.9056	0.0000
6	0.0348	0.9596	0.0056
8	0.0072	0.9556	0.0372
10	0.0016	0.9040	0.0944
11	0.0004	0.8576	0.1420
12	0.0000	0.8136	0.1864
13	0.0000	0.7472	0.2528
16	0.0000	0.5556	0.4444
20	0.0000	0.3500	0.6500
24	0.0000	0.0748	0.9252

Table 2: The twinedness transition. $N = 24$, observation at $t = 90$, 2500 trials per row. Above $k = 12$ the ambiguous outcome is structurally impossible; below it, it is merely rare.

of arcs of length 13 on a ring of 24 is disjoint — checked exhaustively over all 24×24 pairs, with no disjoint pair found — so the outcome is impossible rather than improbable. Monte Carlo cannot see this difference; the structure can.

Remark 27 (The methodological point). Observation 26 is a small instance of the ideal-versus-checkable methodology of [Sci, WG]: the ideal logic *defines* the condition, and the measurement can only ever fail to refute it. An observer that has correctly guessed $k > N/2$ knows something no amount of watching would have told it.

Part II. The swarm: the observer inside

10 Nest-site selection as coalition detection

10.1 The position of this section

Honeybee nest-site selection is among the best-measured collective decisions in biology. Lindauer’s observations and then four decades of work by Seeley, Visscher and collaborators established the mechanism in detail [See10, SB99, SV04], and there is a mature quantitative modelling tradition on top of it: leaky competing accumulators, cross-inhibition dynamics borrowed from models of perceptual decision in primate cortex, and value-sensitive best-of- n analyses [See12, Pai13, Rei17].

Nothing in this section out-predicts those models on trajectories, and it is not offered as a competing dynamical account. What the framework of §§3–9 adds is of a different kind:

- (i) a *logic* in which what a scout is testing is a formula rather than a state variable;
- (ii) a *cost* argument for why she tests that formula and not a different one, which turns an empirical finding into a derivation; and
- (iii) *structural* results — impossibility statements holding for all parameter values — which a statistical model is not in a position to state, and which yield an experiment separating two accounts of what stop signalling is for.

10.2 The mechanism, as measured

Briefly, and only to fix notation. A swarm settles on a branch; some hundreds of its workers scout for cavities. A scout who finds a candidate returns and waggle-dances for it, with

dance vigour proportional to her assessment of the site and decaying by a roughly site-independent decrement on each return trip, so that advocacy for a poor site extinguishes itself while advocacy for a good one is sustained. Uncommitted scouts follow dances and go to inspect. The decision is taken not by agreement among the dancers but by *quorum sensing at the cavity*: when enough scouts are simultaneously present at one site, scouts there return to the cluster and produce worker piping, the cluster warms its flight muscles, and the swarm takes off [SV04, See10]. Scouts committed to one site also deliver *stop signals* — brief vibratory head-butts — preferentially to scouts dancing for other sites, and an analytic model shows this cross inhibition resolves deadlock between equally good alternatives [See12].

Two further facts matter here. First, the quorum threshold is small: figures around fifteen scouts are reported, out of a swarm of ten thousand. Second, and this is the observation that led Seeley to the quorum hypothesis in the first place, swarms occasionally *take off with dissent*: Lindauer and Seeley both recorded launches while two strong coalitions were still advertising distinct sites [See10].

10.3 The mapping

Definition 28 (The swarm as a graded coalition structure). Participants are scouts. A predicate β_c holds of a scout when she is currently committed to cavity c . The namespace $\mathcal{N}$ is the scout pool; the basis of Open is the family of *advocacy sets* — for each cavity c , the scouts that have advocated for c during this episode — closed under union. The graded value of a scout is her dance vigour for the site she advocates. The coalition modality $\langle E \rangle_V^{\mathcal{N}}[\beta_c]$ evaluated by a scout *at cavity* c is the quorum test.

Two features of Definition 28 are worth pausing on, because they are what make the structure a semitopology rather than something simpler.

Observation 29 (Why the meet fails, concretely). Advocacy sets overlap: a scout may dance for A , be stopped or lose interest, and later dance for B . She is in both advocacy sets. But the set of such switchers is not itself an actionable coalition — there is no site it can carry, and no piping it can produce. The intersection of two coalitions is therefore not a coalition, which is exactly the failure of closure under finite intersection that distinguishes a semitopology from a topology. Gabbay’s relaxation is not a technical convenience here; it is a description of what switching bees are.

Observation 30 (The observer is a participant). The scout evaluating the quorum test is a member of the population she is evaluating, and her verdict is what commits the swarm. This is the *resolutive* reading that [GW, ChF] and the herd sections below set aside: consensus among a population resolving the population’s own nondeterminism. The swarm is the first worked instance of it available to us, and the reason it is worth having is that the swarm keeps the two halves apart in a way the theory predicts it must.

Proposition 31 (The swarm exhibits the K/J split). *In the vocabulary of Remark 3, the swarm implements the quantifier and the selection function as distinct behaviours. Dance evaluation — following dances, weighting sites by vigour, being recruited in proportion — is the clause: a map from candidates to values, in the K -monad. Piping and takeoff is the section: a single committed choice, in the J -monad. No bee performs both roles for the same decision, and the transition between them is the quorum test.*

Proposition 31 is a reading rather than a theorem, but it makes a checkable claim: the bees that pipe are the bees that were *at the quorum site*, not the bees that danced most vigorously, which is what the measurements report [VS07].

11 Why quorum and not consensus

Seeley and Visscher framed their experiment as a contest between two hypotheses: *consensus sensing*, in which scouts notice that all dancers now advertise one site, and *quorum sensing*, in which scouts notice that one site is being visited by enough of them. They separated the two experimentally, by delaying quorum formation at the chosen cavity while leaving the rest of the process untouched, and observing that piping and takeoff were delayed accordingly [SV04].

In the present framework these are not two hypotheses about a mechanism. They are two formulae, and they sit at different altitudes.

Definition 32 (The two formulae). Let D be the predicate “is currently dancing” and β_c “advocates c ”. Then

$$\Phi_{\text{quorum}}(c) = \langle E \rangle_V^{\mathcal{N}} [\beta_c], \quad \Phi_{\text{consensus}} = \exists c. \forall q \in \mathcal{N}. ([D](q) \Rightarrow [\beta_c](q)).$$

Proposition 33 (The altitude gap). $\Phi_{\text{quorum}}(c)$ is evaluated by a single $\otimes$ over one open containing the evaluating participant, and is decidable by perception at that open. $\Phi_{\text{consensus}}$ is a universal over the whole namespace with no presented witness, hence not confirmable by any finite perception at a locus; by the modal asymmetry of [Sci, Prop. 6] it is finitely refutable but not finitely confirmable.

Corollary 34 (The mechanism is forced). A participant restricted to the three access grades of [Sci] and to a finite budget can evaluate Φ_{quorum} and cannot evaluate $\Phi_{\text{consensus}}$. If a swarm’s commitment is triggered by a participant’s verdict, it must be triggered by the quorum formula.

This is the framework’s first substantive contact with the data. Seeley and Visscher had to go to Appledore Island to distinguish the two hypotheses; the altitude argument says that only one of them is available to a bee, and would have picked the winner in advance. That the argument is retrospective does not make it vacuous: it is the same argument that will make predictions in §12 and §13, and it is the reason to take those seriously.

Remark 35 (Read versus drive, and where the quorum is sensed). There remains a puzzle in the empirical finding: counting dancers at the cluster would seem *easier* than counting bees at a cavity, since the dancers are all in one place. The read-versus-drive distinction [Play, §5.1] says why the harder channel is the one used. Reading dance vigour at the cluster is perception — cheap, high coverage, and reporting a *lean*. Sensing occupancy at a cavity is interaction, priced by rendezvous — expensive, low coverage, and reporting a *commitment*. The swarm gates its irreversible act on the expensive high-precision instrument and its reversible act, which cavity to go and look at, on the cheap one. That is exactly the allocation Table 5 will exhibit for the pride, arrived at here from the biology rather than from simulation.

12 The $2k$ theorem and what stop signals are for

12.1 A structural bound

Theorem 36 (The $2k$ theorem). Let a swarm have an active scout pool of size S and quorum threshold k , and suppose a scout counts toward the occupancy of at most one cavity at a time. Then two cavities can hold quorum simultaneously only if $S \geq 2k$. In particular, if $S < 2k$ the population cannot present two decided coalitions, for any site qualities, any recruitment and abandonment rates, and any level of cross inhibition.

Proof. Occupancies of distinct cavities are disjoint sets of scouts, so two occupancies of size at least k require at least $2k$ distinct scouts. $\square$

The proof is trivial; the content is in what it is a proof *about*. Theorem 36 is the swarm’s instance of Proposition 23: $S < 2k$ is precisely the condition making the advocacy semi-topology 2-twined, since two advocacy sets of size k in a pool of fewer than $2k$ must meet. Legibility and safety are again one property, and here the property has a number attached to it that can be measured in the field.

Corollary 37 (Takeoff with dissent is a twinedness failure). *The launches with two live coalitions recorded by Lindauer and by Seeley [See10] are possible only in swarms whose active scout pool is at least twice the quorum threshold.*

12.2 The regimes

Theorem 36 divides the parameter space into three, and the division is not about which failure is more likely but about which is *available*.

regime	split	deadlock
$S < k$	impossible	certain
$k \leq S < 2k$	impossible	the failure mode
$S \geq 2k$	possible	displaced by the split

Principle 38 (Cross inhibition changes jobs at $S = 2k$). Below the threshold, the only way the decision can fail is by neither site reaching quorum, so cross inhibition can only be a deadlock breaker. Above it, deadlock is easy to escape and the exposed failure is two quorums, so cross inhibition must be a split preventer. The literature attributes both roles to stop signalling [See12]; the framework says which role is operative is determined by S/k , and that the two are separable experimentally.

12.3 Simulated

We implement the standard structure — independent discovery, recruitment proportional to quality-weighted advocacy, quality-independent abandonment, targeted stop signalling — with the one addition that a site reaching quorum locks its scouts, since piping bees neither abandon nor can be stopped. Two cavities of *equal* quality, the hardest case. A split is recorded when the second site reaches quorum within the piping window; a deadlock when neither reaches it within the horizon. Details in Appendix B.

Observation 39 (What Table 3 shows). The bound of Theorem 36 is necessary and not sufficient: splits are structurally impossible below $S = 30$ and are observed at 0.0130 at exactly $S = 30$, rising to 0.6850 at $S = 50$. The gap between possibility and occurrence is the dynamics, and the framework has nothing to say about its width — that is what the accumulator models are for. What the framework supplies is the floor, and the floor does not move when the rates do.

Observation 40 (The job switch, measured). At $S = 16$, cross inhibition takes deadlock from 0.5120 to 0.0000 and splits are absent from both columns. At $S = 40$, deadlock is absent from both columns and cross inhibition takes splitting from 0.3860 to 0.1170. The same mechanism, doing different work, with the change of duty at $S = 2k$.

12.4 Disparity

Cross inhibition is credited with resolving deadlock *over equal sites* [See12]. The framework predicts its value should therefore fall as the sites become distinguishable, since quality difference then does the symmetry-breaking on its own.

S	no cross inhibition			cross inhibition		
	split	deadlock	clean	split	deadlock	clean
12	0.0000	1.0000	0.0000	0.0000	1.0000	0.0000
16	0.0000	0.5120	0.4880	0.0000	0.0000	1.0000
20	0.0000	0.0770	0.9230	0.0000	0.0000	1.0000
24	0.0000	0.0000	1.0000	0.0000	0.0000	1.0000
28	0.0000	0.0000	1.0000	0.0000	0.0000	1.0000
29	0.0000	0.0000	1.0000	0.0000	0.0000	1.0000
30	0.0130	0.0000	0.9870	0.0010	0.0000	0.9990
31	0.0410	0.0000	0.9590	0.0010	0.0000	0.9990
34	0.1490	0.0000	0.8510	0.0170	0.0000	0.9830
40	0.3860	0.0000	0.6140	0.1170	0.0000	0.8830
50	0.6850	0.0000	0.3150	0.3930	0.0000	0.6070

Table 3: The structural threshold. $k = 15$, so $2k = 30$; the rule above the line is $S < 2k$. Equal-quality cavities, 1000 episodes per cell. Splits are identically zero below the line and first appear exactly at it. Below the line the failure mode is deadlock and cross inhibition abolishes it; above the line the failure mode is splitting and cross inhibition only reduces it. At $S = 12 < k$ no quorum is reachable at all and stop signalling cannot help.

q_B	no cross inhibition		cross inhibition	
	split	best chosen	split	best chosen
1.00	0.4120	0.4898	0.0980	0.5078
0.95	0.3680	0.5870	0.0920	0.5639
0.90	0.3800	0.6758	0.0840	0.6277
0.80	0.3270	0.7845	0.0670	0.7235
0.70	0.2380	0.9003	0.0430	0.8328

The split rate falls with disparity in both columns, as expected, and cross inhibition’s absolute benefit falls with it: 0.3140 at equal quality against 0.1950 at $q_B = 0.7$. There is also a cost visible, and it is worth recording because it is not what one would have guessed: with $S = 40$ and distinguishable sites, cross inhibition makes the swarm choose the *better* site slightly *less* often (0.7235 against 0.7845 at $q_B = 0.8$). It buys decisiveness by suppressing advocacy, and some of what it suppresses is correct advocacy. Whether real swarms show this trade is an open question, and a sharper version of the framework’s claim that resolution is never free.

13 One-sided errors and the encounter-rate prediction

Proposition 16 says a mortal observer conflates a faulty participant with an observation it could not afford to finish. In the swarm the conflation has a concrete form and a testable consequence.

A scout does not count her fellows at a cavity. She registers encounters during a residence spell, and the encounter rate at fixed occupancy falls as the cavity gets larger — more volume, more surface, longer between meetings. Model this as: with true occupancy n , each present scout is registered independently with probability $p = c/a$ where a is effective area, and the verdict is *quorum* when the registered count reaches k .

Proposition 41 (One-sidedness). *Under encounter sensing, false positives are impossible: the registered count never exceeds the true occupancy, so a scout never reports quorum at a site that has not got one. All error is false negative, and the false-negative rate at fixed n increases with a .*

Corollary 42 (The threshold is not a number). *The apparent quorum threshold, measured in true bees present, is $\approx k/p = ka/c$: linear in effective cavity area. A quorum threshold reported as a bee count is a composite of a neural threshold and a geometry, and is not portable between cavities of different size.*

Simulated with $k = 15$ and $c = 0.9$:

effective area	p	apparent threshold	false neg. at $n = k$	false pos.
1.00	0.900	16	0.7951	0.0000
1.25	0.720	21	0.9928	0.0000
1.50	0.600	25	0.9998	0.0000
2.00	0.450	33	1.0000	0.0000

The apparent threshold is 16, 21, 25, 33 against areas 1, 1.25, 1.5 and 2: ratios of 16.0, 16.8, 16.7 and 16.5, linear to within the granularity of the measurement. False positives are identically zero, as they must be.

Remark 43 (The experiment). Corollary 42 is falsifiable with the apparatus already in use. Nest boxes of differing internal volume, at matched quality on every other dimension, should show apparent quorum thresholds — counts of scouts present at the moment piping begins — scaling with volume. If instead the count is invariant, bees are not encounter-sensing and Proposition 41 is wrong about the mechanism. Since cavity volume is itself a quality cue for bees, the manipulation has to be designed against that confound; matched-volume boxes differing in internal partitioning would be one route.

14 What to measure

Collecting the swarm predictions in the form an experimentalist would want them.

- E1. **The split floor.** Estimate the active scout pool S and the quorum threshold k for each swarm. Predict: takeoff with dissent occurs only in swarms with $S \geq 2k$, with no exceptions, and independently of site quality. A single well-documented split in a swarm with $S < 2k$ refutes Theorem 36’s applicability and therefore the advocacy-set reading of Open.
- E2. **The job switch.** With stop signalling suppressed or reduced — pharmacologically, or by using swarms with experimentally reduced scout pools — predict that the failure mode changes character at $S = 2k$: deadlock below, splitting above. This separates the deadlock account of stop signalling from the split account, which the current literature does not distinguish because both are true in different regimes.
- E3. **Geometry-dependent thresholds.** Per Remark 43.
- E4. **One-sidedness.** Predict that scouts err by failing to detect present quorums and never by detecting absent ones, so that measured piping onsets lag true quorum attainment with a distribution having no left tail.

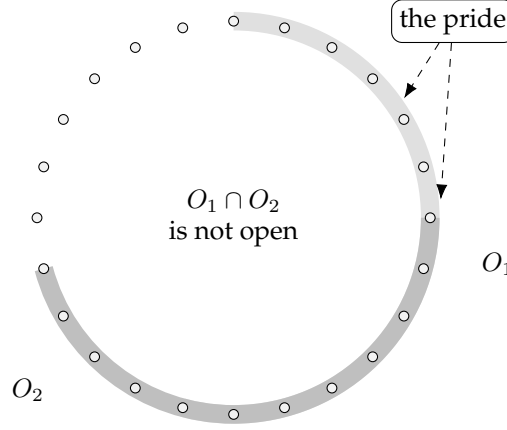


Figure 1: The herd as a semitopology. Arcs of length at least k are the basis of Open; their union is open, their intersection in general is not. The pride observes participants one at a time, paying for each.

- E5. **The margin clock.** Predict that the advocacy margin between the two leading sites at an undecided moment predicts the wait to quorum monotonically. Measured in simulation at 59.7, 35.1, 18.4, 8.4 ticks across margin quartiles (Table 8); the corresponding quantity in the field is dancer counts per site, which is already routinely recorded.
- E6. **Switchers are not a coalition.** Per Observation 29, scouts who have advocated both sites should be over-represented among bees that neither pipe nor stop-signal. If switchers pipe at the same rate as loyalists, the advocacy-set semitopology is the wrong Open and should be replaced by the site-occupancy one.

E6 is the one that most directly tests the *logic* rather than the arithmetic, since it asks whether the structure the bees act on is the one the framework says it is.

Part III. The pride: the observer outside

15 The worked example: a pride and a herd

15.1 The population

$N = 24$ animals on a ring. Animal i carries a heading $s_i \in \{-1, 0, +1\}$ — break left, undecided, break right — and deposits a confidence $g_i \in [0, 1]$ recording how firmly its own neighbourhood agrees with it. The dynamics is asynchronous majority over a window of radius $W = 2$ with per-update noise $\varepsilon = 0.03$: at each tick a uniformly chosen animal adopts the sign of the coupling-weighted sum of headings in its window. Couplings are 1 except at designated *seams*, where they are 0.

A stampede begins with one startled animal and runs for 400 ticks. The herd has *reached a conclusion* when the final total heading exceeds half the herd in magnitude; runs that fragment are discarded from the accuracy measurements and counted separately in §16.

15.2 The pride's ladder

Seven hypotheses, priced on the schedule $\kappa(d) = 2^d$ per candidate examined [Play]: a heading costs 2 (spatial, depth one), a deposited confidence costs 4 (spatial, depth two), testing

rung	coverage	precision	accuracy	cost	return
R0 $\top$	1.0000	0.4985	0.4985	0.0	—
R1 one heading	1.0000	0.9002	0.9002	2.0	+0.20087
R5 topen member	0.8498	0.9744	0.9031	4.0	+0.00144
R2 sample of five	1.0000	0.9868	0.9868	10.0	+0.01394
R3 crisp $\exists\Box$	0.7510	1.0000	0.8755	190.2	−0.00062
R4 graded $\langle E \rangle_V^N$	0.7510	1.0000	0.8755	192.0	+0.00000
R6 ν drive forward	1.0000	0.9832	0.9832	1344.0	+0.00009

Table 4: The ladder, ordered by cost. $k = 13$, observation at $t = 90$, 4000 stampedes that reached a conclusion. *Coverage* is the fraction of trials on which the rung issues a verdict at all; *precision* is its accuracy given that it does; *accuracy* folds the two together by tossing a coin on b , which is what an animal that must commit actually does. *Return* is the marginal gain in accuracy per metered unit against the next cheaper rung.

one open costs a further 4 (the modal step), and driving the population forward one tick costs 8. Observations are cached within a single evaluation, so no participant is billed twice for one verdict.

R0. $\top$. No theory; commit at random.

R1. One heading. Read a single animal.

R2. Sample of five. Read five animals; take the sign of the sum.

R3. Crisp $\exists\Box$. Is there an actionable coalition uniformly heading one way? Since a union of basis arcs is uniform iff each constituent is, it suffices to test the N arcs of length exactly k .

R4. Graded $\langle E \rangle_V^N$ over the quorum algebra of Definition 13, returning value and witness.

R5. Topen member. Read one animal of a known topen and its confidence; predict the whole topen.

R6. ν : drive the population forward six sweeps and then apply R4. This is the fixed point of Remark 10 evaluated by simulation instead of by inspection.

R0–R2 and R5 are *perception*: they read structure the herd has already computed. R3–R4 are perception at modal depth one. R6 is *driving*. The read-versus-drive distinction of [Play] is thus laid out along the ladder, and the pricing schedule is what makes the comparison mean anything.

15.3 The ladder measured

Read naively, Table 4 says the coalition rungs are a bad purchase: R3 has *negative* marginal return, costs nineteen times what R2 costs, and has lower folded accuracy. This replicates the headline of [Play] — yield, not truth — and if that were the end of it the note would be reporting that Coalition Logic is ecologically worthless.

It is not the end of it. The folded accuracy column is the wrong instrument.

t	sample coverage	sample precision	coalition coverage	coalition precision
30	1.0000	0.9160	0.0037	1.0000
50	1.0000	0.9640	0.1263	1.0000
70	1.0000	0.9797	0.4740	1.0000
90	1.0000	0.9837	0.7527	1.0000
120	1.0000	0.9873	0.9223	1.0000

Table 5: The coalition modality trades coverage for precision. 3000 stampedes; $k = 13$. Across all five observation times the coalition rung issued 6839 verdicts and got every one of them right.

15.4 Lean versus commitment

R2 and R3 answer different questions. A five-animal sample reports which way the herd is *leaning*. The coalition modality reports that it has *committed*. A lean can reverse; a decided coalition, under twinedness, cannot.

Observation 44 (What the modality buys). Over 6839 coalition verdicts, precision was 1.0000 at every observation time, including at $t = 30$ where coverage is 0.37%. The sample majority, which always answers, was wrong on 8.4% of its verdicts at $t = 30$ and on 1.3% at $t = 120$.

So the coalition rung is not a worse predictor of direction; it is a different instrument. It converts a probabilistic lean into a categorical commitment, paying for the conversion in coverage and in tokens. For a pride the distinction is the whole game: a charge committed on a lean that then reverses is worse than no charge at all, because the pride has spent its ambush.

Remark 45 (The verdict is not a theorem). Precision 1.0000 is measured, not proved. With $\varepsilon > 0$ the dynamics can in principle unwind a decided coalition, and nothing in Proposition 23 forbids it: twinedness rules out two *simultaneous* conclusions, not a later reversal of one. What the measurement shows is that under this dynamics the event did not occur in 6839 opportunities. A general statement would need a hypothesis relating the noise rate to the coalition size, and is left open (Q61).

Principle 46 (A second currency). The yield metric of [Play] — accuracy gained per metered unit — prices instruments that always answer. It systematically underprices instruments whose value is that they sometimes decline to. Coverage and precision must be reported separately, and an ecology in which commitment is irreversible will pay for precision at prices the yield metric calls irrational.

15.5 Earliness

R6, which drives the population forward, is the earliest instrument by a wide margin: it is at 0.938 when every perceptual rung is near chance. This is what a lookahead buys, and it is also why it costs 1344 units against R2's 10. The ecological reading of Table 6 is that simulation is a luxury good: it is genuinely the best instrument on the board, and almost nothing can afford it. The identity of the R3 and R4 rows is Remark 15 made visible.

rung	$t=20$	40	60	80	100	140	first ≥ 0.80
R0 $\top$	0.486	0.492	0.512	0.519	0.509	0.492	—
R1 one heading	0.623	0.705	0.818	0.868	0.915	0.949	60
R2 sample of five	0.845	0.948	0.975	0.986	0.983	0.981	20
R3 crisp $\exists\Box$	0.500	0.518	0.657	0.813	0.920	0.974	80
R4 graded	0.500	0.518	0.657	0.813	0.920	0.974	80
R5 topen member	0.587	0.693	0.791	0.861	0.914	0.951	70
R6 ν	0.938	0.966	0.983	0.981	0.985	0.987	10

Table 6: Folded accuracy against observation time, and the earliest sampled time at which each rung reaches 0.80. 1200 stampedes.

seams	Pr(herd concludes)	pride coverage	pride precision
0	0.7257	0.7520	0.9993
1	0.6780	0.6030	1.0000
2	0.4984	0.3645	0.9986
3	0.2957	0.3015	1.0000
4	0.4495	0.0910	1.0000
5	0.2877	0.0685	1.0000
6	0.2331	0.0115	1.0000

Table 7: Cohesion against legibility. $k = 13$, observation at $t = 90$. As the herd fragments its own quorum structure, the pride’s coverage collapses by a factor of 65 while its precision does not move.

16 The counter-move

Nothing so far has given the herd a say. But a population that is being read has an interest in not being readable, and Principle 25 tells it exactly where to attack: break quorum intersection.

In the model this is done by inserting seams — bonds of zero coupling — into the ring. A herd with enough seams cannot form an arc of length k that is internally uniform, because every such arc crosses a seam and the two sides of a seam decide independently.

Observation 47 (Illegibility is denial of liveness, not induction of error). Across Table 7 the pride’s precision stays within $[0.9986, 1.0000]$ while its coverage falls from 0.7520 to 0.0115. The herd does not make the pride wrong. It makes the pride silent.

This is exactly the signature of a consensus failure seen from the far side. An adversary who breaks quorum intersection in Paxos does not cause two correct replicas to decide differently — that is what the intersection property is there to prevent, and the property is not what the adversary can touch. The adversary causes the protocol to *stop deciding*. Safety is preserved, liveness is lost. Read from outside, the same event is an observer that keeps being right and stops being able to say anything.

Corollary 48 (The price of illegibility). *The herd pays for the pride’s silence in its own coordination: across the same rows, the probability that the herd reaches any conclusion at all falls from 0.7257 to 0.2331. A herd that cannot be read is a herd that cannot decide which way to run.*

Principle 49 (The legibility trade). For a population under observation, the safety–liveness trade-off of consensus and the cohesion–illegibility trade-off of predation are one trade-off.

A population buys privacy with the same currency it buys agreement, and cannot spend it twice.

Principle 49 is the note’s second payoff from the reversal, and the one that seems most likely to generalise beyond herds. It is a claim about any population whose members must coordinate while being observed by something that profits from predicting them, which is a wide class.

Remark 50 (Dual use). Read as engineering rather than ecology, Principle 49 says that a protocol cannot be made unobservable by weakening its quorum structure without losing exactly the liveness that quorum structure was providing. Privacy mechanisms that work by fragmenting agreement are paying for it in the protocol’s own progress, and the exchange rate is the one in Table 7.

17 Compression: what a topen is worth

A topen is a transitive open: an open T such that any two opens meeting T meet each other. Gabbay’s theory makes toopens the pivotal notion because their members are guaranteed to decide alike. For an observer this is a compression result.

Principle 51 (One observation buys n predictions). If T is a topen and $q \in T$, then observing q determines the observer’s prediction for every member of T . The value of the observation is multiplied by $|T|$, and the observer’s optimal strategy is to spend on *identifying toopens* rather than on observing members.

Measured in a two-cluster herd (seams at positions 0 and $N/2$, so the two half-rings decide independently and each is a topen):

	agreement with the observed animal	predictions per observation
same cluster	0.8683	6.5
across the seam	0.5606	5.9

Inside the topen a single observation yields 6.5 predictions at 86.8% accuracy — 4.81 correct net of errors. Across the seam the same observation yields nothing above chance. The entire value of the observation is concentrated in the topen, and an observer that has misidentified the topen boundary is paying for information it is not receiving.

Remark 52 (Why topen identification is the expensive part). Identifying a topen is a question about *pairs* of opens, so its evaluation is quadratic in the basis where checking a single coalition is linear. This is the observer’s version of the fact that transitivity is the expensive axiom. It is also where the ultrametric confinement of [Sci, Prop. Confinement] bites hardest: a topen boundary in the wrong place is a shape error, and shape errors are not reachable by refinement.

18 The margin is a clock

Remark 15 left the grading looking useless: at $k = 13$ it returns exactly what the crisp modality returns. The resolution is that the observer should not evaluate the graded formula at the coalition size it cares about. It should evaluate a *relaxed* formula — coalitions of size $k_{\text{warn}} = 7$ — purely as an early-warning instrument, while retaining $k = 13$ as the definition of commitment.

Conditioning on stampedes that have *not* committed at $t = 60$, and banding by the tropical margin of the relaxed formula:

margin quartile	mean margin	ticks to commitment	direction correct
lowest	0.0864	42.4	0.6080
second	0.7648	29.6	1.0000
third	0.8000	28.9	1.0000
highest	0.9872	18.0	1.0000

Table 8: The relaxed margin as an early-warning signal. 1500 stampedes uncommitted at $t = 60$, $k = 13$, $k_{\text{warn}} = 7$.

Observation 53 (What grading buys). The margin is monotone in both columns that matter: time to commitment falls from 42.4 to 18.0 ticks across the quartiles, and directional correctness rises from 0.6080 to 1.0000. A crisp verdict at $k = 13$ says only “not yet”. The relaxed graded value says how long “not yet” will last.

For a pride this is the difference between knowing that the herd is undecided and knowing that it has eighteen ticks to get into position. Grading does not improve the answer to *which way*; it answers *when*, which is a question the crisp logic cannot ask.

The swarm shows the same shape, and there the quantity is one field workers already record. Conditioning on episodes undecided at $t = 250$ with a mild quality disparity, and banding by the advocacy margin between the two sites:

margin quartile	mean margin	ticks to quorum
lowest	0.1016	59.7
second	0.3009	35.1
third	0.4625	18.4
highest	0.5914	8.4

A sevenfold spread in the wait, read off a quantity that costs nothing extra to observe. For a swarm this is the difference between a cluster that should be warming up and one that should not; for a pride it is the difference between getting into position and wasting an ambush.

Principle 54 (Two dials). The value algebra and the coalition size must be varied together. Grading at the operative coalition size is crisp in disguise (Remark 15); relaxing without grading gives an unreliable crisp verdict (the lowest quartile of Table 8 is right only 60.8% of the time); relaxing *and* grading gives a graded verdict about a weaker proposition, which is the useful object.

Principle 54 connects the two lattices that have been running in parallel throughout: the relaxation lattice of [Sci], which orders hypotheses by logical strength, and the algebra table of [GW, ChF], which orders resolutions by what they can express. They are not independent knobs, and a note that turned only one of them would have concluded that grading buys nothing.

19 Altitude: where this sits in the choice hierarchy

The formulae of this note are clauses, and [ChF] assigns clauses a choice type χ — a bound on the Weihrauch degree of the resolver a clause demands. It is worth locating the coalition modality on that lattice, because doing so predicts what a real observer can afford.

Proposition 55 (The coalition modality is a width engine). *Over a presented basis, $\langle E \rangle_V^N$ is a finite $\oplus$ and carries choice type $\perp$: its value is computed by inspection with no oracle. When the basis is a family indexed by a countable presented set, the modality is the width engine of [ChF, §5], of degree $C_{\mathbb{N}}$ when the index is $\mathbb{N}$ and $C_{2^{\mathbb{N}}} \equiv \text{WKL}$ when the index is a powerset.*

Proposition 56 (Coherence is a depth engine). $\Phi_{\text{coh}} = \nu X. \langle E \rangle_V^N([\beta] \otimes \langle K \rangle X)$ is a greatest fixed point over the run tree and is therefore the depth engine, of degree WKL ; its content is compactness.

Corollary 57 (Exact herd-reading is expensive). *Predicting that a population will continue to agree combines both engines, which are Weihrauch-incomparable [ChF, §5], and so sits at $C_{\mathbb{N}^{\mathbb{N}}}$ or above.*

Remark 58 (And so real prides do not do it). Corollary 57 is not merely a complexity remark; combined with the pricing schedule it is a prediction about behaviour, and Table 4 bears it out. R6, the only rung that evaluates anything resembling Φ_{coh} , costs 1344 units against R2’s 10. The instruments that survive an ecological budget are the ones that read structure the herd has already deposited — headings, confidences, arcs — which is the read-versus-drive thesis of [Play] arriving from the direction of degree theory rather than of measurement. A pride does not simulate zebra. It looks at their necks.

Remark 59 (Strong fairness, again). The identification of ν -over-the-run-tree with WKL is the same identification that gave strong fairness its degree in [ChF], where the content of the weak/strong fairness gap turned out to be compactness. The present note adds a third instance of the same pattern: the gap between “some coalition agrees now” and “some coalition keeps agreeing” is the same gap.

19.1 Subject reduction, and the same suspect

Question 60 (Subject reduction for coalition clauses). Does a shard whose clauses mention $\langle E \rangle_V^N$ have invariant choice types under reduction?

The answer is expected to be no, for the reason it was no in [ChF]. Reduction substitutes a quoted process for a name and $*y$ drops an arbitrary received process into term position, so a clause whose declared coalition structure is finitely presented can, after one communication, come to mention a received process carrying a ν over a non-presented Open. Strength is mailed in. The repair is the same: a stratified *where*, declaring a ceiling and checking levels on drops. Reflection is now the suspect in a fourth place, and Krivine’s observation that DC is realised by adding a `quote` instruction to the machine remains the sharpest statement of why: the rho calculus has had its choice primitive in the syntax since [Mer05].

20 What is shown and what is not

Shown. That Coalition Logic admits an observational reading which needs only its model theory (Proposition 4); that the participant/process-structure distinction of [Cons, §7.1] collapses under reflection (Proposition 6); that the correct value algebra for a quorum is min-based and that this is the fragment where the observer receives a witness (Propositions 12, 14); that a mortal observer conflates fault with poverty (Proposition 16, Table 1); that quorum intersection is what makes a hypothesis about a population well posed (Proposition 23);

and, by measurement in one model, that the coalition modality trades coverage for precision (Table 5), that a population's counter-move costs the observer liveness rather than accuracy (Table 7), that topen membership is where the value of an observation is concentrated (§17), and that the relaxed graded margin is an early-warning clock (Table 8).

From the swarm: that quorum sensing rather than consensus sensing is forced by the access grades (Proposition 33, Corollary 34); that two simultaneous quorums require $2k$ scouts and hence that takeoff with dissent is confined to swarms with $S \geq 2k$ (Theorem 36); that cross inhibition's duty changes at that same threshold (Principle 38, Observation 40); and that under encounter sensing quorum errors are one-sided with the apparent threshold scaling in cavity geometry (Proposition 41, Corollary 42).

Not shown. That any swarm prediction has been *tested*; E1–E6 of §14 are proposals, and the simulated tables show only that the predictions are non-vacuous in a model with the right mechanisms in it. That the advocacy-set reading of Open is the right one rather than the site-occupancy reading — E6 is precisely the experiment that would tell them apart, and Theorem 36 holds under either. That the framework improves on the accumulator models in any respect they address; it does not, and §10 says so at the outset. That precision 1.0000 for the coalition modality is anything but an artefact of this dynamics at this noise rate (Q61). That the pricing schedule $\kappa(d) = 2^d$ is the right one — it is inherited from [Play], where it was already flagged as a choice, and the qualitative shape of Table 4 is robust to varying it while the boundaries are not. That a ring with arcs is a good model of a herd; it is a good model of a semitopology that is not a topology, which is what was needed. That the observer's guessed Open can be learned efficiently — the note assumes a guess and prices its use, and says nothing about acquiring it beyond the negative result inherited from ultrametric confinement. That any of this extends to n -twinedness for $n > 2$, which is where heterogeneous quorum systems live and where the interesting engineering is.

Deliberately not attempted. The resolute reading — using a population's consensus to settle a race inside the observer — is not developed here. Remark 3 says why it is the harder direction: it requires a section where the clause provides only a weighting. It remains an interesting direction and is the natural next note.

21 Open problems

Question 61 (Irreversibility). Give a condition on the noise rate, the coalition size and the neighbourhood structure under which a decided coalition is stable with probability one, so that Observation 44 becomes a theorem rather than a measurement.

Question 62 (Learning Open). Principle 19 says the semitopology is the hypothesis. Give a revision procedure over guessed semitopologies, priced, and locate it on the ultrametric: which errors in Open are refinement-reachable and which are shape errors requiring recombination?

Question 63 (Heterogeneous quorums). Extend the worked example from 2-twinedness on a ring to genuinely heterogeneous quorum systems, where different participants have different notions of an actionable coalition. This is where Gabbay's theory has its real content and where Corollary 24 becomes a statement about *whose* question is well posed.

Question 64 (The compatibility relation as an observation). Semiframes replace intersection by a compatibility relation $*$, read in [Real] as co-realizability: φ and ψ share a witness. What does an observer pay to test $A * B$, and is the compatibility relation cheaper to observe than

the coalitions themselves? If so, an observer should work in the semiframe rather than the semitopology.

Question 65 (The cost of cross inhibition). Table in §12 suggests cross inhibition buys decisiveness partly by suppressing correct advocacy, choosing the better site slightly less often at moderate disparity. Is this an artefact of the implementation, and if not, what is the swarm’s optimal σ as a function of S/k and site disparity?

Question 66 (Which Open do the bees act on?). Observation 29 takes the basis of Open to be advocacy sets, which makes switchers the reason the meet fails. Site occupancies would give a different and nearly-degenerate structure. E6 of §14 distinguishes them empirically; is there an argument from cost that picks one in advance, as Corollary 34 picks the quorum formula?

Question 67 (Composition of observers). Two prides watching one herd are two learners with overlapping namespaces in the sense of [Comp]. The competitive dilution result there says each retains a share of the deep inquiry it could afford alone. Does watching a population deplete it as a source, and if so is the herd a rate-limited source or a prey item in the trophic sense of [Sci, §10]?

Question 68 (Stochastic evaluation). Section 7.2 of [Cons] proposes expressing Coalition Logic axioms as MeTTaIL rewrite rules and using the weighted-GSLT machinery of [WG] to explore probabilistic consensus. The graded modality of Definition 9 is a natural key for a weight map. Does the partition discipline of [WG, §3] survive keys mentioning $\langle E \rangle_V^N$, given that requirement R3 there restricts keys to structural formulae?

Question 69 (The second currency). Principle 46 asserts that coverage and precision must be priced separately. Give the ecology in which precision is worth its cost: a game in which a reversed commitment is strictly worse than no commitment, and compute the exchange rate at which R3 invades a population of R2 players.

22 Conclusion

The companion note asked how to build a coalition from a proof. This one asks how to notice that one has formed. The machinery is the same — Gabbay’s modality, the rho calculus, the join — but running it in the observational direction changes what is hard and what is easy. The proof theory that blocks synthesis is not needed; the model theory that exists suffices. What becomes hard instead is everything the synthesis reading takes for granted: which sub-populations can act as one, whether the observer can afford to find out, and whether a verdict of “don’t know” is a fact about the world or about the purse.

The two examples are not two illustrations of one point. The swarm is the case where the observer is a participant and its verdict is the decision, so the clause and the resolver appear together and the theory has to say how they are kept apart; the swarm keeps them apart by putting them in different bees. The pride is the case where the observer has no purchase at all, so the theory has to say what watching alone can buy; the answer is irreversibility, bought with coverage, at a price the ordinary yield metric calls irrational. Neither case would have produced the other’s result.

Three things came out of the work that were not put in. The first is that quorum intersection, which consensus theory keeps for safety, is what makes a population legible from outside at all — and in the swarm it acquires a number, $S \geq 2k$, that says which swarms are able to disagree with themselves. The second is that a population resists being read by denying coverage rather than by inducing error, and pays for the resistance in its own

capacity to agree, so that privacy and coordination are drawn from one account. The third is smaller and more uncomfortable: an observer poor enough sees a world full of broken things, because a participant it could not afford to examine and a participant that is genuinely stuck return the same value. In the swarm that has a shape — scouts should fail to see quorums that are there, and never see quorums that are not — and it means the number a field biologist writes down as the quorum threshold is partly a fact about the bee and partly a fact about the box. That last point is not about bees. It is about anything that has to pay to look.

A The pride and the herd in rholang

The listing below is the population and the observer as rho processes, in the variant of rholang carrying *where*-clauses. Names are manufactured by quotation rather than bound by *new* where the freshness argument of [Sci, §3] applies; *new* is retained where it reads better and is sugar. Guards written in the ideal logic rather than the implemented fragment are marked `//! ideal`.

```
new herd, pride, tick in {

  // ---- a participant
  -----
  // animal k lives at the manufactured name @[*herd, k]; it holds its
  // heading as a resting send and re-deposits it after each update.

  contract @[*herd, "animal"](@k) = {
    for (@h <- @[*herd, k]) {
      // read the window; couplings are data on the bond channels
      for (@w <- @[*herd, "window", k]) {
        match w {
          tot => {
            // the deposit: heading AND the confidence the
            neighbourhood
            // has already computed. This is what makes perception
            cheap.
            @[*herd, k]!(if (tot > 0) 1 else if (tot < 0) -1 else h) |
            @[*herd, "conf", k]!(agree(w) / total(w))
          }
        }
      }
    }
  }

  // ---- an open set
  -----
  // an arc of length K starting at a is presented as a name predicate;
  // testing it is a join over its members, which is the coalition
  // modality's term-level shadow read in the observation direction.

  contract @[*pride, "arcUniform"](@a, @K, @d, ret) = {
    new acc in {
      acc!(true, 0) |
      for (@ok, @i <= acc) {
        if (i == K) { ret!(ok) }
        else {
          for (@h <- @[*herd, (a + i) % 24]) { // peek, not consume
```

```

        acc!(ok and (h == d), i + 1)
    }
}
}
} |

// ---- the crisp coalition modality
-----
// exists O in Open. p in O and all q in O have heading d.
// A union of basis arcs is uniform iff each arc is, so the N arcs of
// length exactly K exhaust the test.

contract @[*pride, "Ebox"](@d, ret) = {
  new scan, hit in {
    scan!(0) |
    for (@a <= scan) {
      if (a == 24) { ret!(false) }
      else {
        new r in {
          @[*pride, "arcUniform"]!(a, 13, d, *r) |
          for (@u <- r) { if (u) { ret!(true) } else { scan!(a + 1) } }
        }
      }
    }
  }
} |

// ---- the graded coalition modality
-----
// <E>_V psi = max over opens O of min over q in O of psi(q).
// The witness arc is returned alongside the value: Prop 12 says the
// value is attained, and the pride charges at the witness, not the
// value.

contract @[*pride, "gradedE"](@d, ret) = {
  new scan in {
    scan!(0, 0.0, -1) |
    for (@a, @best, @arg <= scan) {
      if (a == 24) { ret!(best, arg) }
      else {
        new m in {
          @[*pride, "arcMin"]!(a, 13, d, *m) |
          for (@v <- m) {
            if (v > best) { scan!(a + 1, v, a) } else { scan!(a + 1,
best, arg) }
          }
        }
      }
    }
  }
} |

// ---- the observer
-----

```

```

// the hypothesis sits in the where-clause. The pride commits when
// the
// graded value clears zero, and charges at the witness arc.

for (@t <- tick) {
  new v in {
    @[*pride, "gradedE"]!(1, *v) |
    for (@val, @arc <- v where val > 0.0) {      //! ideal: <E>_N psi
      @[*pride, "commit"]!("right", arc)
    }
  }
}
}

```

Two conventions are worth flagging. The peek $\llcorner$ rather than the linear receive $\leftarrow$ is what makes the scan *perception* rather than *interaction*: a linear receive would consume the animal's heading, which is to say the pride would move the zebra by looking at it. The separation is the same one forced in [Play, §4.3], where threat markers had to live on their own names lest perception rendezvous with the move channel and play the move. Grade one and grade two of access are kept apart by choice of namespace, and that is the only mechanism available.

B The computations

All numbers are produced by `swarm.py` and `herd.py` (standard library only, single seed 20260813 in each). Run `python3 swarm.py` and `python3 herd.py`.

Parameters. $N = 24$ participants; window radius $W = 2$; per-update noise $\varepsilon = 0.03$; horizon 400 ticks; seam coupling 0, elsewhere 1. A stampede counts as concluding when the final total heading exceeds $N/2$ in magnitude.

Pricing. $\kappa(d) = 2^d$ per candidate examined: a heading costs 2, a deposited confidence costs 4, a tested open costs a further 4, and one tick of driving costs 8. Observations are cached within a verdict.

The swarm: `swarm.py`

Parameters. S scouts, two cavities, quorum threshold $k = 15$; per-tick independent discovery $\nu = 0.0060$, recruitment coefficient $\rho = 0.0500$ applied to $q_i D_i / S$, quality-independent abandonment $\alpha = 0.0050$, cross-inhibition strength $\sigma = 0.0200$ when enabled; piping window 60 ticks, horizon 1500 ticks, 1000 episodes per cell, seed 20260813. A site reaching quorum locks its scouts, since piping bees neither abandon nor can be stopped; this is what makes the counting argument of Theorem 36 bind on the simulation as well as on the biology.

The measurements.

1. `structural_threshold` — split, deadlock and clean-decision rates against S , with and without cross inhibition, equal cavities (Table 3).
2. `unequal_sites` — the same against site disparity at $S = 40$ (§12).

3. `one_sided_errors` — false negative and false positive rates under encounter sensing, and the apparent threshold in true bees, against effective cavity area (§13).
4. `margin_clock` — the advocacy margin as a predictor of the wait to quorum (§18).

What the simulation is and is not. It is a standard discovery–recruitment–abandonment–inhibition scheme, chosen so that the mechanisms named in the text are present and none other is. It is not calibrated against field data and its rate constants are not estimates of anything. Its only jobs are to confirm that the structural floor of Theorem 36 is respected, to show that the predictions of §12 and §13 are non-vacuous, and to give the regime table numbers.

The herd: `herd.py`

The measurements.

1. `ladder` — coverage, precision, folded accuracy and mean cost per rung; 4000 concluding stampedes (Table 4).
2. `reversal` — coverage and precision of the sample and coalition verdicts at five observation times; 3000 stampedes (Table 5).
3. `twinedness_witness` — the constructed split state deciding both directions at $k = 12$, and the exhaustive check over all 24×24 arc pairs that no two arcs of length 13 are disjoint (Observation 26).
4. `earliness` — folded accuracy against observation time; 1200 stampedes (Table 6).
5. `twinedness` — the ambiguity transition in k ; 2500 stampedes per row (Table 2).
6. `cohesion_vs_legibility` — conclusion probability, observer coverage and observer precision against seam count (Table 7).
7. `budget_confusion` — the two bottoms; 2500 stampedes (Table 1).
8. `topen_multiplier` — agreement inside and across a seam, unconditioned on the herd concluding; 3000 stampedes (§17).
9. `margin_predicts_commitment` — the relaxed margin as a clock; 1500 stampedes uncommitted at $t = 60$ (Table 8).

A negative result worth keeping. The first version of measurement 9 evaluated the graded modality at the operative coalition size $k = 13$ and returned a margin of exactly zero on every trial, because min over thirteen participants is zeroed by one dissenter. That failure is Remark 15, and it is the reason Principle 54 is stated: the note would have concluded that grading buys nothing had the coalition size not been relaxed alongside the algebra.

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